·

( )

2026-06-13

I. — ,

II. — · · · 7

III. · — 200

IV. —

. · . .

- · **end-to-end**
- 12 , **200 + SCI**
- · ·

- GWAS, PRS (PRScs · PRScsx), scRNA-seq
- Nanostring spatial transcriptomics
- Proteomics · lipidomics · methylation
- xMWAS

22 +

R

200+

SCI !!!

8

R&D

— 5 + 7 (· ·)

—

- GWAS **2~8%**
- (CYP2C19) **~12%**
- ‘ (state)’ → ‘ (process)’

- — MR, pQTL/eQTL
- —
- — ·

(DNA → RNA → Protein → Metabolite)

- **Genomics / Epigenomics**
- **Transcriptomics** — scRNA-seq, spatial
- **Proteomics** — Olink PEA
- **Metabolomics / Lipidomics**

— MOFA · xMWAS · Mendelian Randomization · QTL · EWAS

: vs

Single-omics GWAS / DEG / EWAS

-log10(P) Effect size IL6 CRP

n = 1,248Case-control

FDR < 0.0517 hits

Top pathwayInflammation

Feature Effect Adj.P

IL6 beta = +0.42 0.004

CRP beta = +0.31 0.011

CXCL8 beta = -0.27 0.019

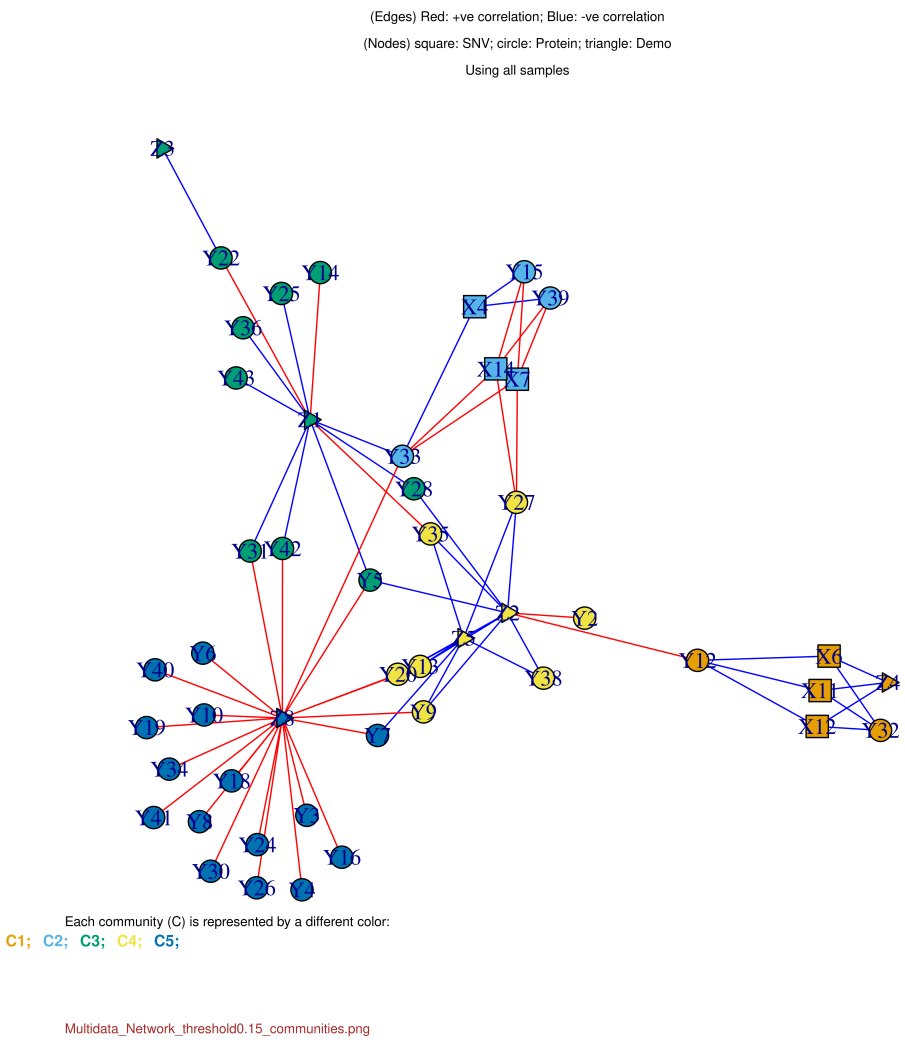
“ ” , .

Integrated

Genomics Transcriptomics Proteomics Metabolomics Outcome

CohortBOMICS / Stroke / Validation

FilterResponder · FDR < 0.05 · Module 3



Multidata_Network_threshold0.15_communities.png

CommunityAPOC1-CRP module

Latent factorInflammation axis

InteractiveLayer-on/off selection

5

STEP 1 · , , / /

STEP 2 · , baseline ,

STEP 3 · · ·

STEP 4 · / /

STEP 5 · / / /

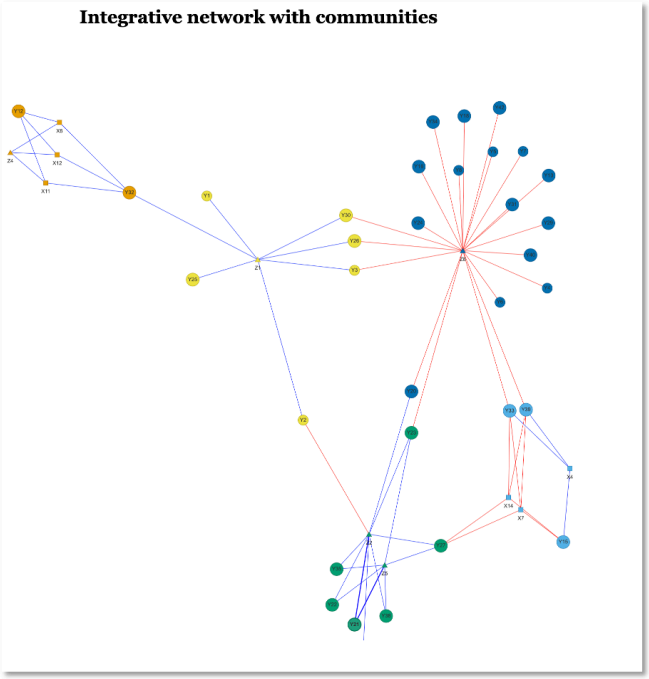
— (· ·) , · · .

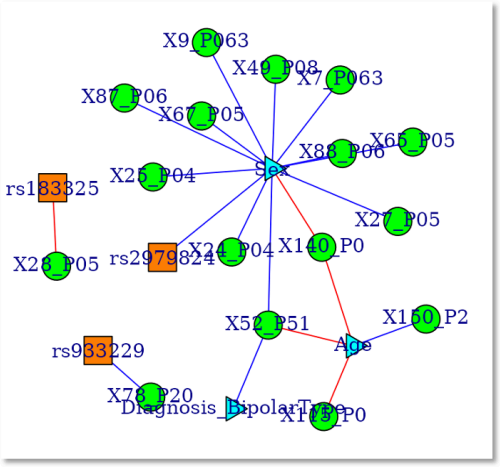
Case 1.

- , , proteomics, SNP, PRS .
- Lithium Valproate , .

- · , feature .
- Table 1, linear/logistic regression , SNP-level GWAS, network analysis, LASSO, Random Forest .

- , SNP- - , , `[75% train / 25% test]` `[10-fold CV]` .





Logistic regression predicting Sex

	crude OR.(95%CI)	crude P value	adj. OR.(95%CI)	adj. P value
H_MDE_H1.2: 1 vs 0	0.958 (0.433,2.119)	0.915	1.007 (0.327,3.098)	0.9909
H_MDE_H1.1: 1 vs 0	0.914 (0.309,2.706)	0.8708	0.768 (0.153,3.852)	0.748
H_MDE_H1.7: 1 vs 0	0.801 (0.495,1.297)	0.3677	0.72 (0.432,1.201)	0.2088
H_MDE_H1.4: 1 vs 0	1.794 (1.068,3.013)	0.0273	2.02 (1.13,3.612)	0.0177
H_MDE_H1.3: 1 vs 0	0.864 (0.531,1.406)	0.5554	1.166 (0.662,2.054)	0.5949
H_MDE_H1.0: 1 vs 0	0.989 (0.408,2.396)	0.9809	0.915 (0.262,3.197)	0.8887
No. of observations	302			
AIC value	390.54878			

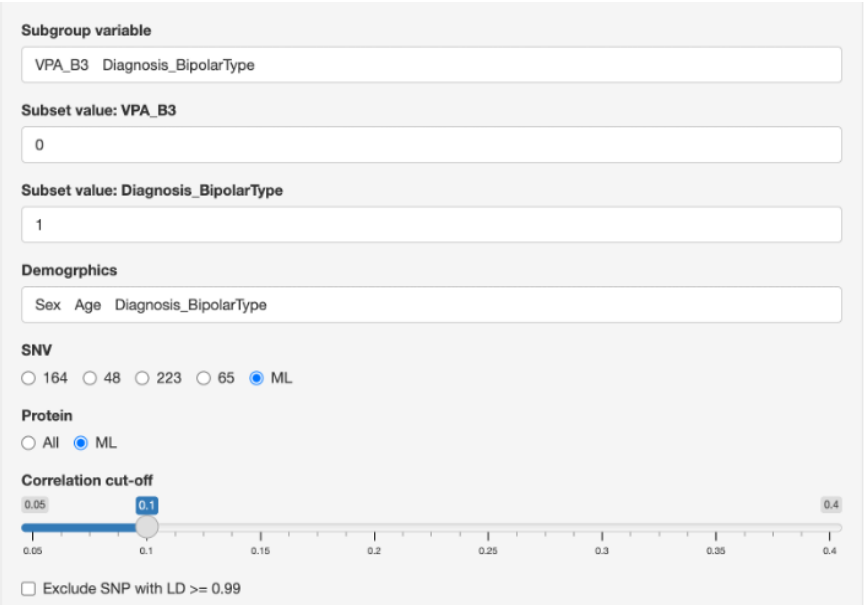
Case 1.

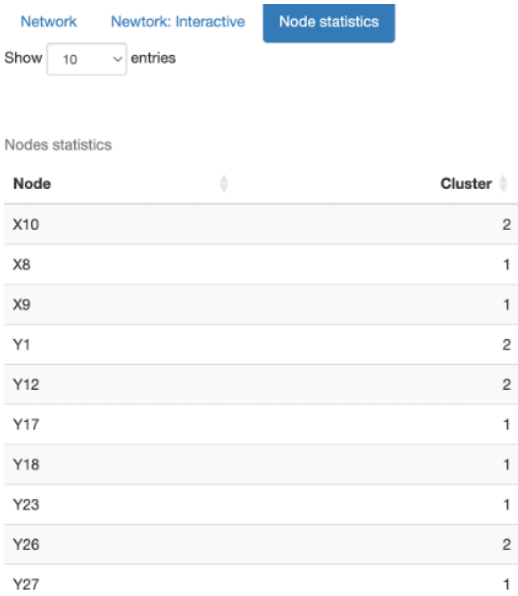
- PRS summary statistics, phenotype, eQTL/mQTL, Brain-mMeta .
- score , lead SNP post-GWAS .

- ancestry-informed PRS , SNP- .

- PRScs, PRScsx trait score logistic ROC , SMR/HEIDI SNP .

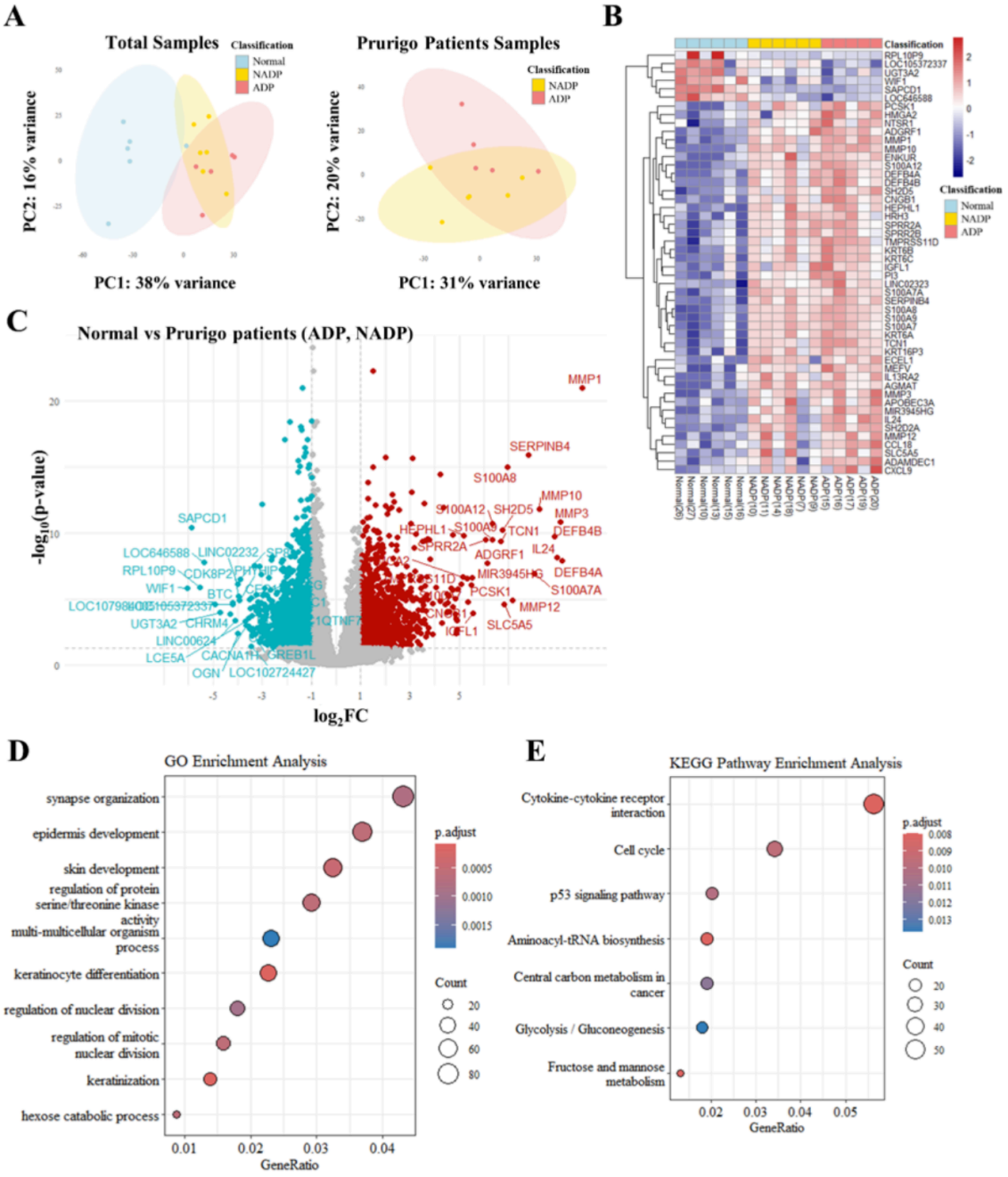
- PRS SNP , SLC25A12 gene-based SMR `p_SMR = 0.381` SNP .





Node	Cluster
X10	2
X8	1
X9	1
Y1	2
Y12	2
Y17	1
Y18	1
Y23	1
Y26	2
Y27	1

Case 2. -

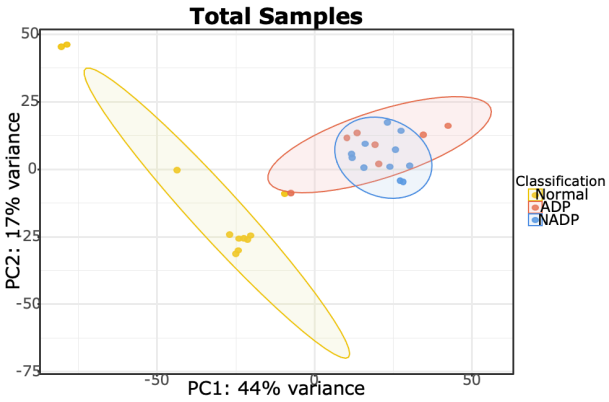

A
Total Samples
Classification
Normal
NADP
ADP
PC2: 16% variance
PC1: 38% variance
Prurigo Patients Samples
Classification
NADP
ADP
PC2: 20% variance
PC1: 31% variance
B
Classification
Normal
NADP
ADP
C
Normal vs Prurigo patients (ADP, NADP)
-log10(p-value)
log2FC
MMP1
SERPINB4
S100A8
MMP10
S100A12
SH2D5
MMP3
DEFB4B
TCN1
IL24
ADGRF1
DEFB4A
S100A7A
MMP12
SLC5A5
SAPCD1
LOC646588
RPL10P9
WIF1
UGT3A2
LINC00624
LCE5A
OGN
LOC102724427
D
GO Enrichment Analysis
synapse organization
epidermis development
skin development
regulation of protein serine/threonine kinase activity
multi-multicellular organism process
keratinocyte differentiation
regulation of nuclear division
regulation of mitotic nuclear division
keratinization
hexose catabolic process
GeneRatio
p.adjust
Count
E
KEGG Pathway Enrichment Analysis
Cytokine-cytokine receptor interaction
Cell cycle
p53 signaling pathway
Aminoacyl-tRNA biosynthesis
Central carbon metabolism in cancer
Glycolysis / Gluconeogenesis
Fructose and mannose metabolism
GeneRatio
p.adjust
Count

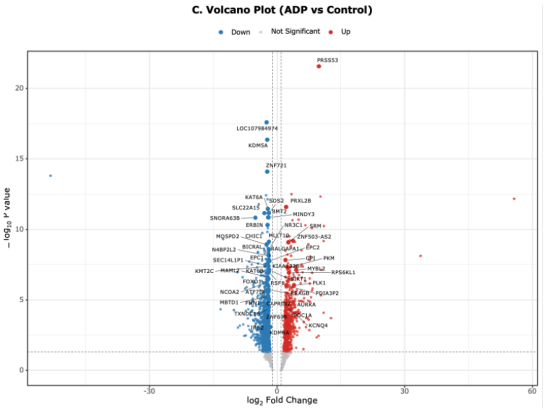
- lipidomics, A/B/C site-specific .
- sensitive skin, prurigo , lipid .

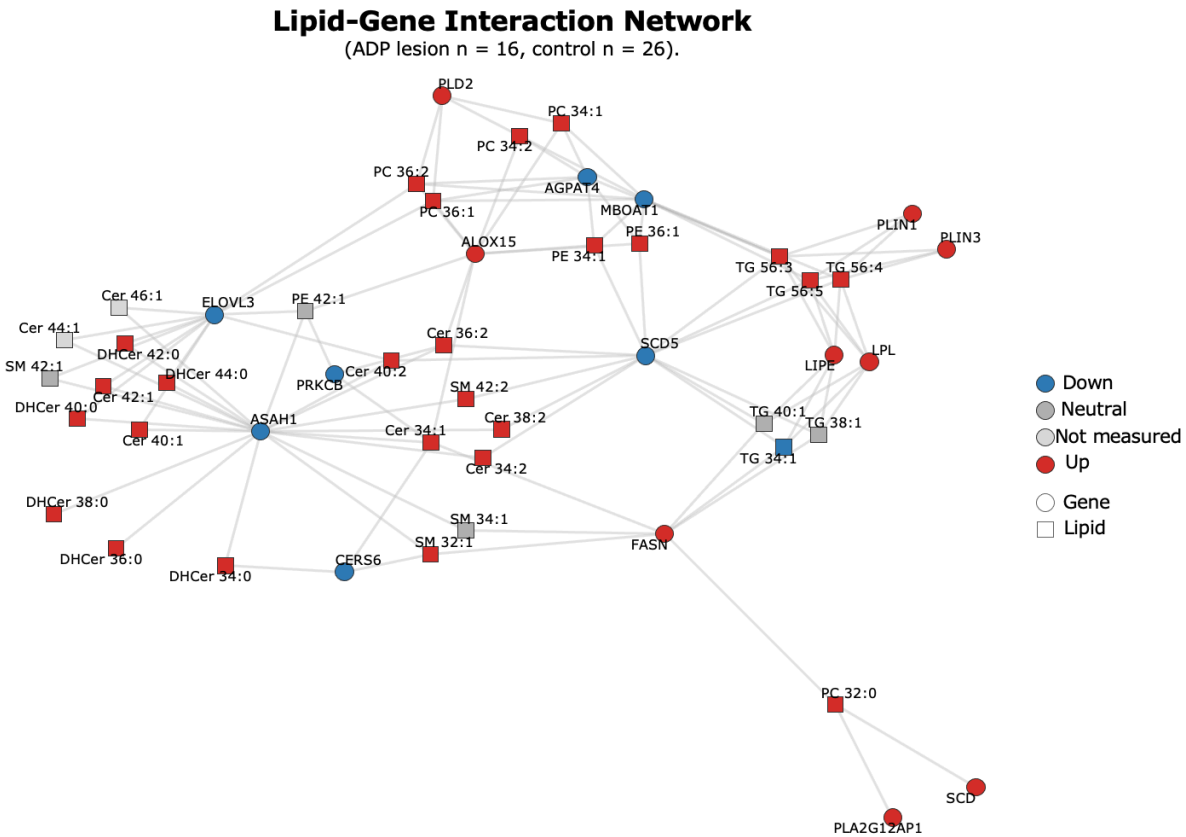
- lipid axis , signature .
- fold-change limma + empirical Bayes , BH-FDR, Spearman correlation, PCA/UMAP .

- lipid fold-change Figure , 3 unique lipid 248 18 species .

Case 2. -







- transcriptome 29 , lipid-matched subset, · multiomics .
- lipid matched analysis .

- lipidomics transcriptome , · pathway multiomics .
- DESeq2 DEG VST PCA, GO/KEGG enrichment, BH-corrected correlation lipid-gene .

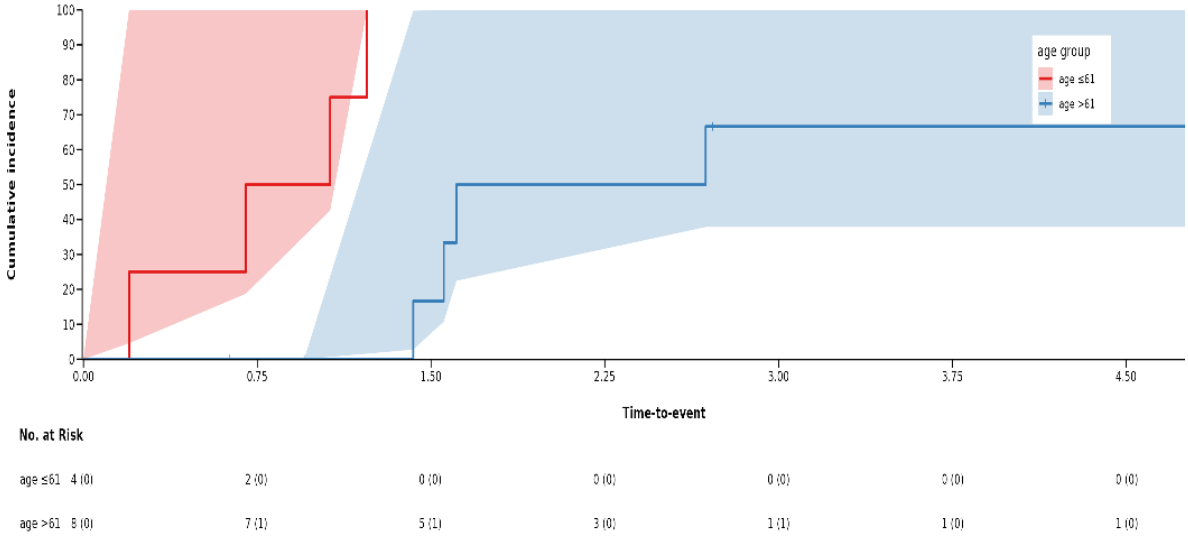
- Figure 1~5 multiomics correlation , `LAD vs Healthy` `[FDR \< 0.05]` 3 .

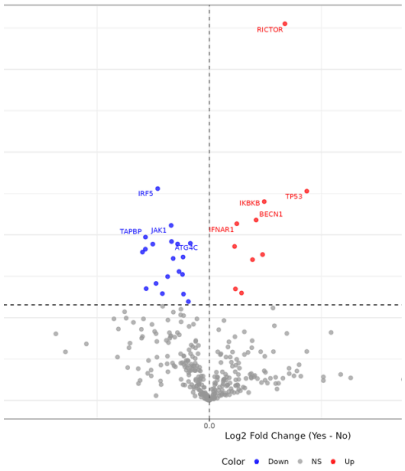
Case 3. STING1-lysosome

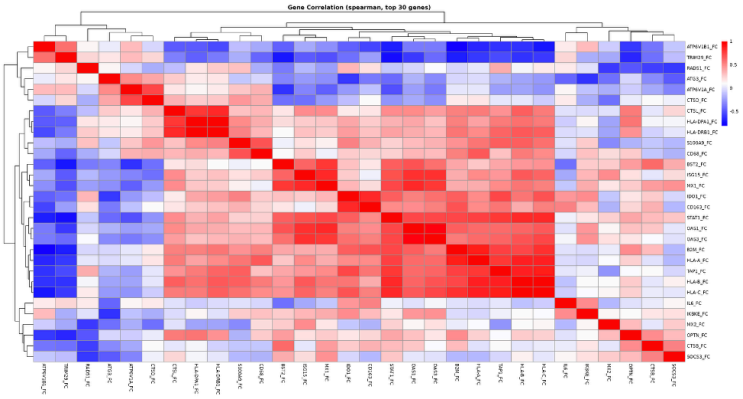
- GeoMx spatial transcriptomics TCGA/GTEx · .
- , STING1-lysosome 245 .

- STING1-lysosome , .
- Kaplan-Meier, Cox , DESeq2, GO/KEGG/PPI, eigengene correlation, paired Wilcoxon .

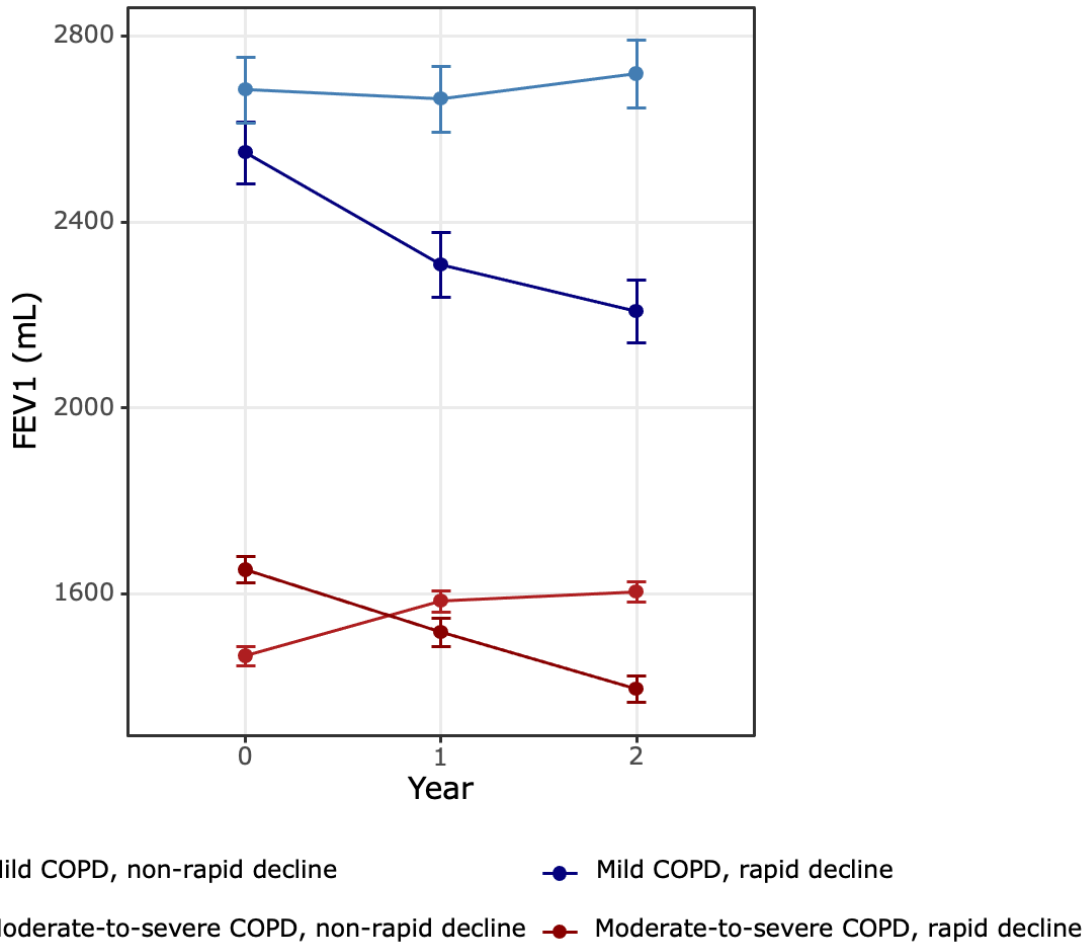
- , volcano, module network, eigengene correlation , PFS event OS .

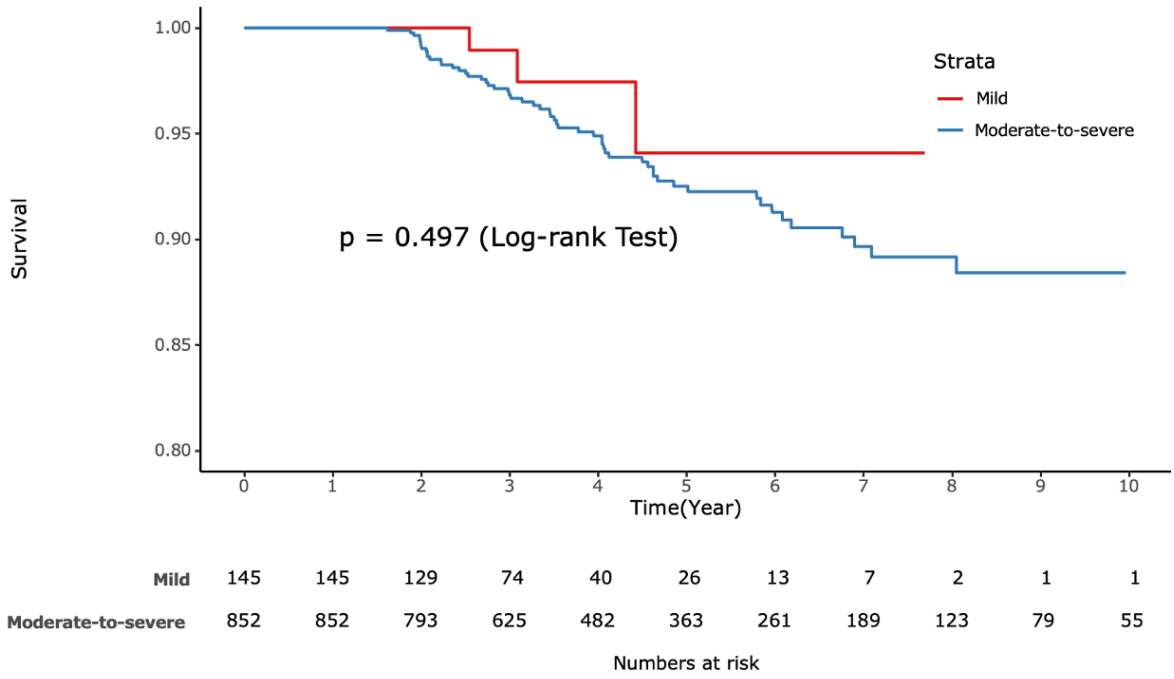



RICTOR
IRF5
TAPBP
JAK1
RTOR
IKBKB
BECN1
TP53
IFNAR1
0.0
Log2 Fold Change (Yes - No)
Color
Down
NS
Up


Gene Correlation (spearman, top 30 genes)
1
0.5
0
-0.5

Case 4. FEV1 COPD





- KOCOSS COPD , baseline FEV1 annual decline 4 longitudinal .
- trajectory , .

- rapid decline , , .
- linear mixed model FEV1 , logistic regression negative binomial , Kaplan-Meier Cox .

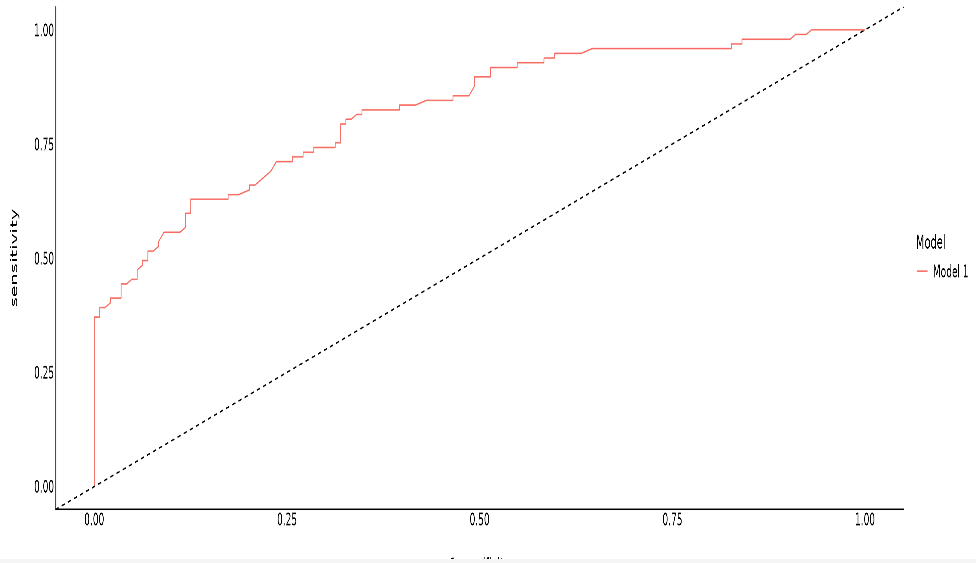
- 4 trajectory Figure , mod-severe rapid decline Y1 `OR 1.69 (1.03-2.77)`, Y1+2 `OR 2.41 (1.02-5.73)` , `HR 1.08 (0.63-1.83)` .

Case 5.

- methylation marker MPR panel, test set .
- marker control percentile feature .

- diagnostic marker , .
- ROC/AUC, logistic regression percentile thresholding, 10-fold CV, LASSO, Random Forest , .

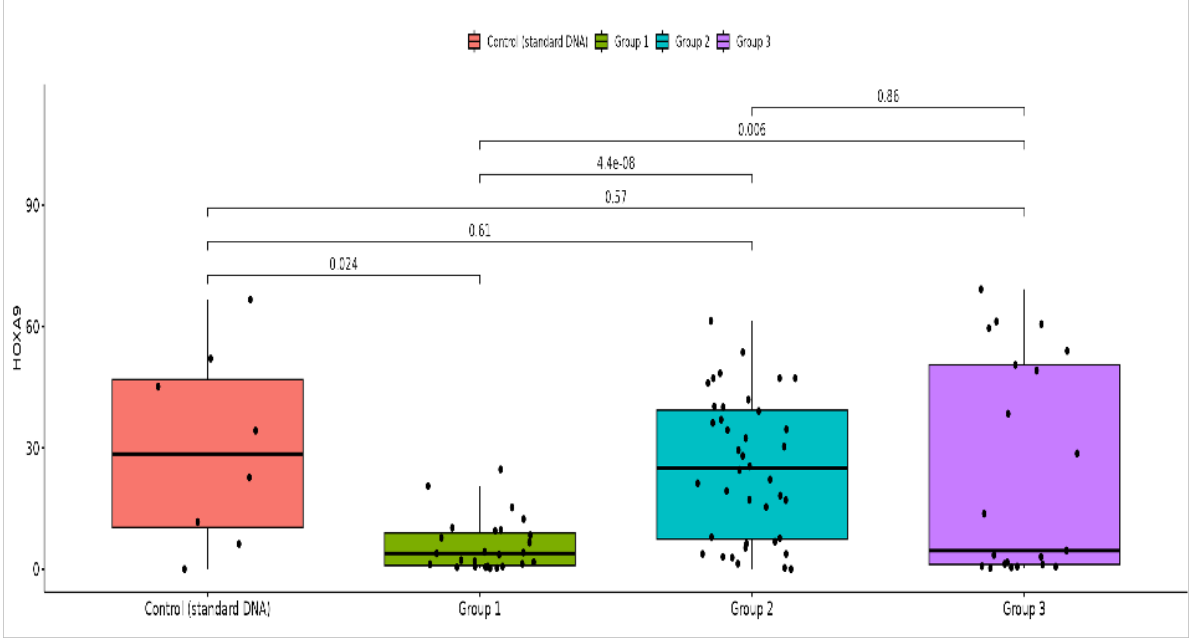
- ROC , cutoff , , AUC best-threshold, sensitivity/specificity, PPV/NPV, 95% `sensitivity` `specificity` .



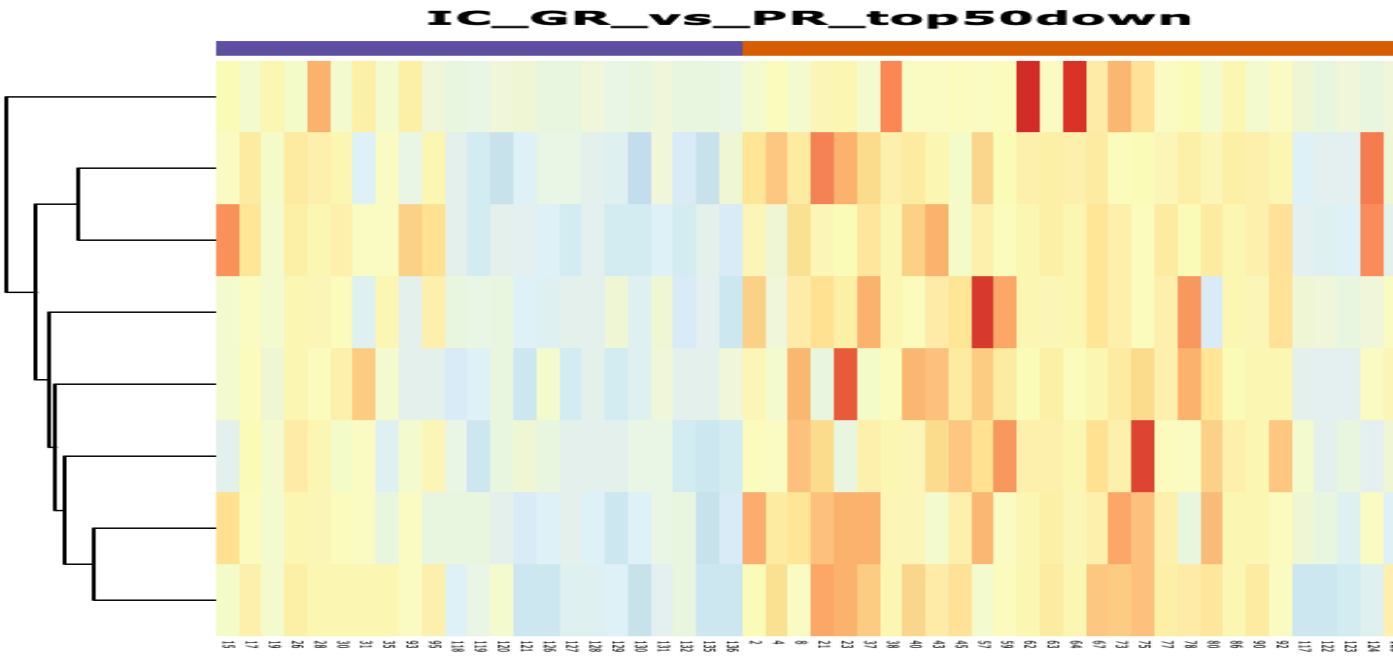
Independent variables for Model 1

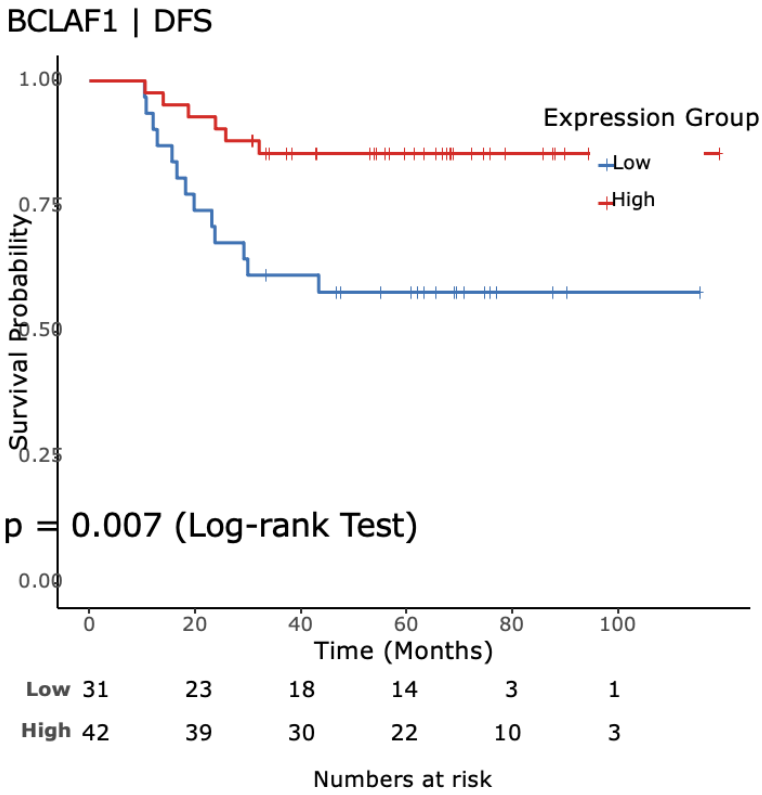
Sex | Smoking_cut3 | MPR_score3 | MPR_score1 | MPR_score10_cut1 | Age

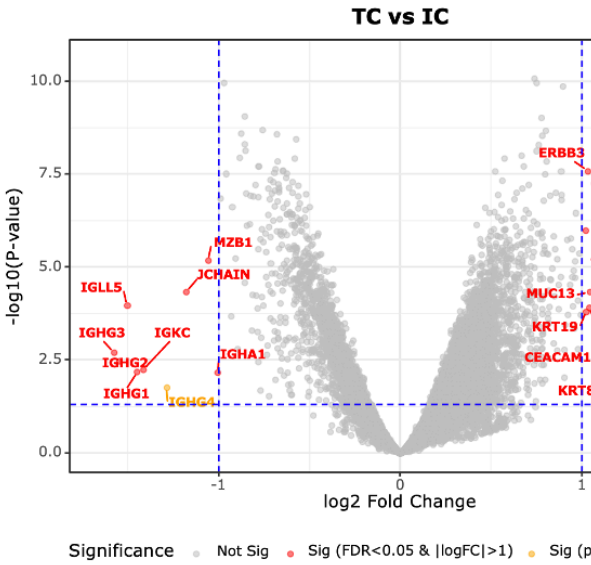
Prediction Model	AUC	95% CI
Model 1	0.825	0.771-0.879


Control (standard DNA)
Group 1
Group 2
Group 3
0.86
0.006
4.4e-08
0.57
0.61
0.024
HOXA9
90
60
30
0
Control (standard DNA)
Group 1
Group 2
Group 3

Case 6.


IC_GR_vs_PR_top50down





- CCRT TC/IC spatial transcriptomics .
- , discovery validation .

- DEG , .
- limma + eBayes DEG , heatmap/volcano , maxstat + Kaplan-Meier, ROC/Youden .

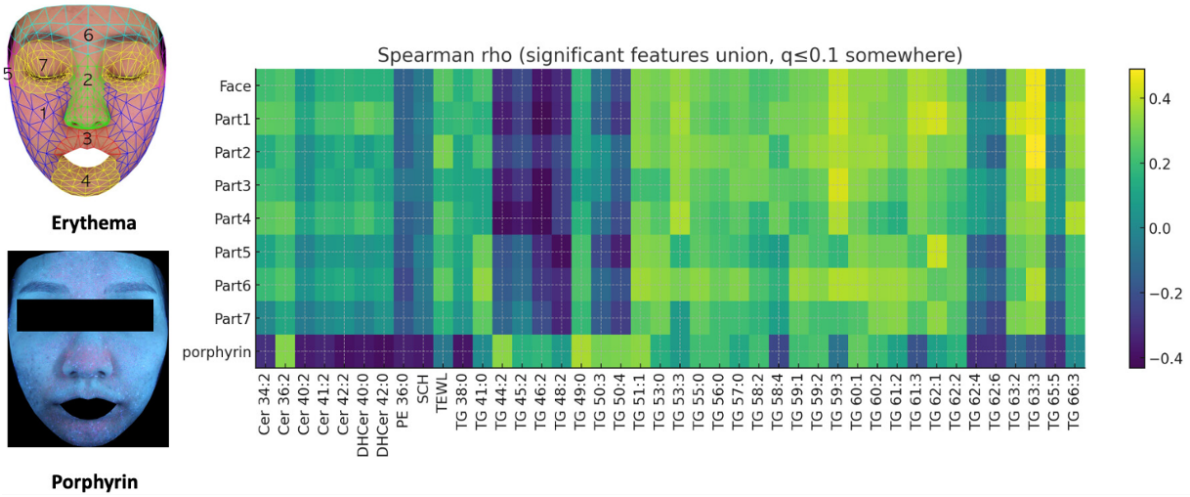
- DEG · Figure , 69 (61), binary validation 1,071 .

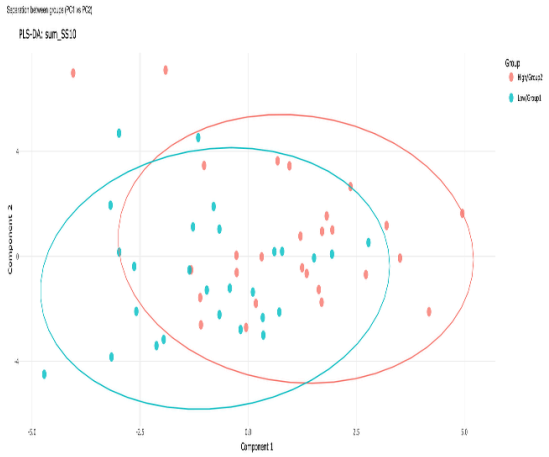
Case 7.

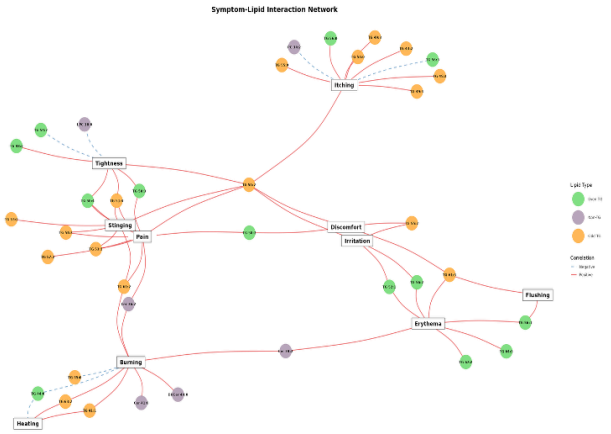
- TEWL, SCH, erythema, porphyrin, SS-10, lipid species .
- high-dimensional , p-value .

- · lipid pattern , symptom domain exploratory .
- Spearman correlation linear regression 1 , PCA, PLS-DA, PERMANOVA, network analysis, LASSO .

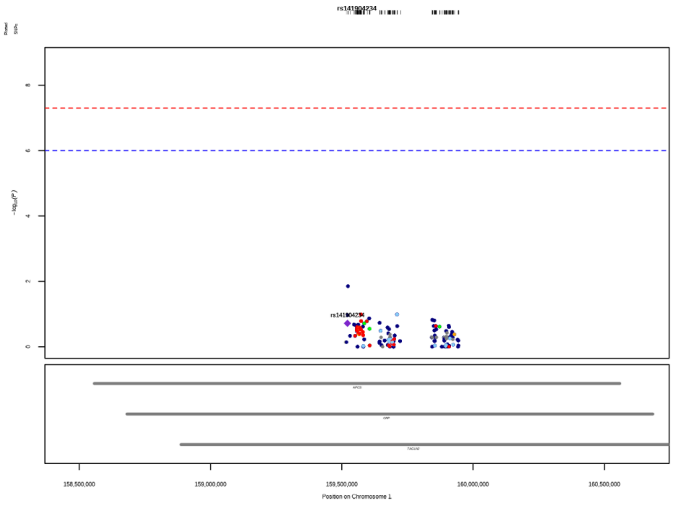
- correlation heatmap symptom-lipid network , SS-10 item 112 lipid `FDR q \< 0.05` lipid , patient-control PLS-DA `PERMANOVA P = 0.753, R² = 0.007` exploratory layer .

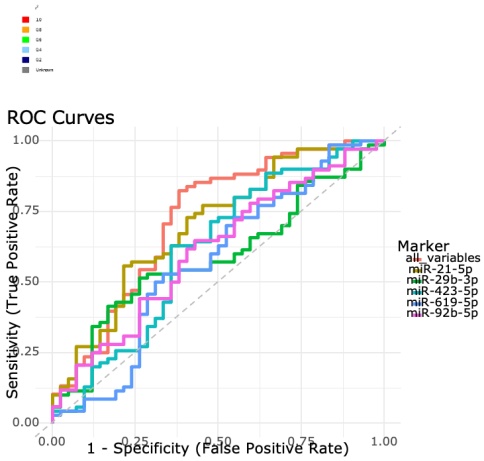


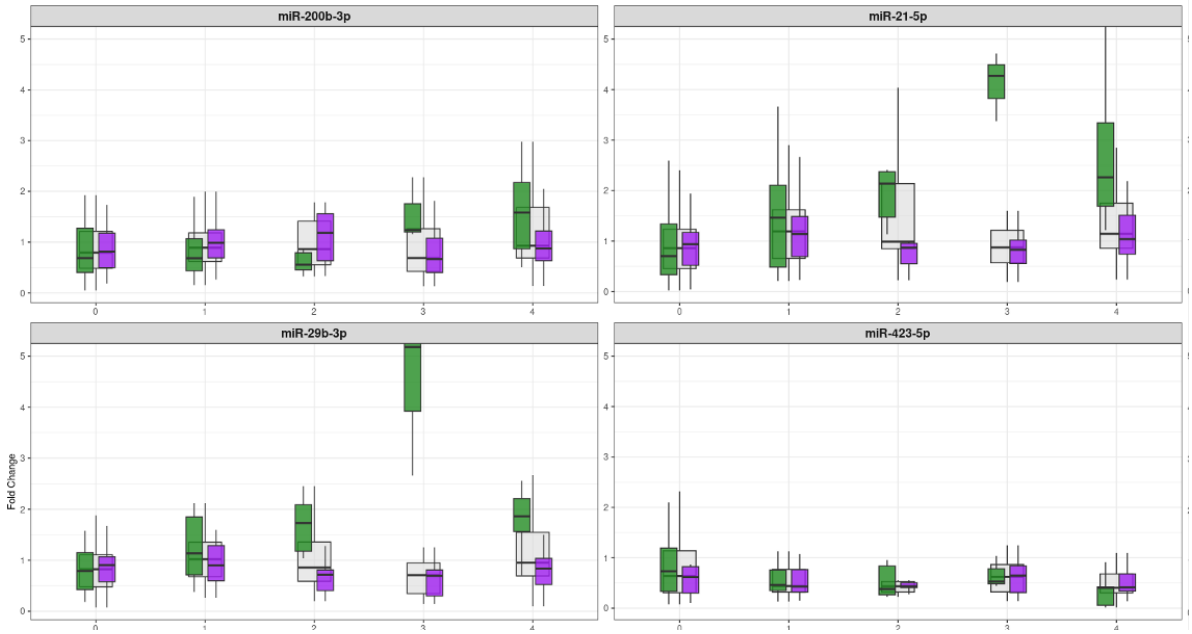




Case 8. qPCR miRNA







- qPCR miRNA, , , , .
- , marker .

- miRNA marker panel , screening .
- qPCR Table 1, boxplot + pairwise test, logistic ROC/AUC, UpSet combination search, LASSO, Random Forest .

- 8 miRNA $3 \cdot 4 \cdot 5$-marker ROC Figure PPT · , AUC · · raw qPCR/pins .

Case 9. · — 200

(Neurology, 2025.04)

- 40~60 **200**
- MetS HR **1.24** (95% CI 1.19~1.30)

- Vascular dementia HR 1.21, Alzheimer 1.12
- **NEJM Journal Watch** ,

- **data.table**
- Multivariable Cox + subgroup interaction
- 1~5 **dose-response**

  (J Am Heart Assoc, 2025)

- CRCS-K +
- ΔHR elevated
- HIRA **(MPR)**

  5 — HIRA · · · outcome

  — jsmodule Shiny

( )

jsmodule — tb1module2, logisticModule2, coxModule, kaplanModule, **subgroupModule**, rocModule, **timerocModule**, ggpairsModule2, jsPropensityGadget.

Stroke subtype GWAS

- TOAST × genotyping array QC
- 1000G EAS imputation, regenie
- GIGASTROKE · STROMICS
- Manhattan / QQ Shiny

(Clopidogrel · Statin · NOAC)

- **CYP2C19 star allele** (PharmVar)
- SLCO1B1 · ABCG2 statin PK
- CPIC level 1A
- × PGS interaction

(MOFA · MR)

- scRNA-seq × Olink × genotyping MOFA
- Mendelian randomization (IVW, Egger)

- pQTL/eQTL → druggable target

HIRA

- / /
- IPTW · time-varying Cox (tmerge)
- Forestploter plot
- · ·

—

“ ” , · · . 7

- — (GWAS, scRNA-seq)
- —
- — ·
- — SCI

- ·
- R (Git)
- Shiny
- · (flextable / rvg PPT)
-

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